PSYCON: STUDYING CONTEXTUAL COMMUNICATION BEHAVIOR THROUGH EVIDENCE-LINKED PSYCHOLOGIST OBSERVATIONS

BEHAVIORAL AND SOCIAL SCIENCES RESEARCH WORKING DRAFT

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ABSTRACT

PSYCON implements an evidence-linked workflow for studying contextual communication behavior using audio, speaker-specific measurements and psychologist-provided reference ratings. The instrument normalizes recordings, diarizes speakers, transcribes speech, preserves timestamped exchanges and connects observable measurements to reviewable reports. Its source-score pilot contains 423 ratings from forty participant-session examples across nine recordings and five known connected groups. On the common 402-rating cohort, compact nested development improved exact agreement from 57.21% to 62.19% and reduced mean absolute error from 0.562 to 0.490. Training-fold median and mode references achieved 64.43% and 65.67% agreement, respectively, defining the comparison that future behavioral prediction must exceed. A 51-configuration pipeline audit reproduced saved results, compared feature families under training-only preprocessing and inner grouped selection, and identified a temporal-ordering defect corrected with regression tests. All nine normalized recordings preserved source audio duration within 15 milliseconds. The resulting research contribution is an executable, reproducible pipeline with evidence provenance, explicit missing-data handling and auditable model comparisons. Results are retrospective development; class imbalance, unresolved identities and unreviewed labels require independent validation before claims of reliable behavioral prediction or communication benefit.

Keywords behavioral observation · social interaction · communication · psychologist ratings · measurement validity · reproducibility

1 Behavioral question and significance

Two participants can have similar speaking time yet behave differently in a discussion. One may answer a direct question, another may respond to a different point; either may speak briefly because of turn allocation rather than disengagement. The behavioral problem is to describe what a person did in an identifiable exchange without turning a recording into a permanent judgment of that person.

PSYCON examines this problem through contextual psychologist observations linked to conversational evidence. Its principal question is: *To what extent can speaker-specific talking patterns and discussion context predict contextual A–T ratings reviewed by school psychologists on held-out participants and recordings, and does adding strictly earlier personal history improve the support and usefulness of communication reports?*

The scientific object is behavior in context. Audio processing and supervised prediction are measurement tools. The current project is positioned in Behavioral and Social Sciences, with an emphasis on social interaction and operational behavioral observation. It is not a study of whether a software interface is attractive or whether a device can transmit packets. The Society for Science science-project criteria emphasize a testable question, appropriate variables and controls, systematic analysis, sufficient evidence, and understanding of limitations [1].

The potential benefit is a reviewable account of participation, reciprocity, contribution structure, and responses to challenge. A participant or psychologist could inspect the exact moment supporting an observation, correct attribution,

and see uncertainty. That benefit remains prospective. No study in the reviewed artifacts establishes improved well-being, learning, communication skill, or clinical outcomes.

2 Psychological background and research gap

2.1 Interaction is sequential and context-dependent

Conversation involves the organization of speaker turns, responses, and opportunities to enter. Foundational conversation-analysis work examines this organization rather than interpreting speaking duration as a complete description of social behavior [2]. PSYCON uses that distinction operationally: a timing overlap is measurable, but deciding whether it was an intentional interruption requires the surrounding exchange and human review.

A response after disagreement is similarly ambiguous. Reduced participation could reflect being interrupted, a topic change, moderator allocation, language access, fatigue, or a deliberate decision to listen. The project therefore asks observers to describe an event, earlier behavior, a response, its duration, and its effect on the exchange. It does not infer an emotion from a pause or assign a trait from one session.

These operational choices do not constitute an established psychological theory of all communication. They narrow the question to behavior the source recording can support. A scientifically useful observation must be attributable to the correct person, defined reproducibly, and interpreted within an opportunity to exhibit it.

2.2 Measurement reliability and construct validity

Construct validity concerns whether evidence supports the interpretation of a measure, rather than whether a form merely has a psychological label [3]. The A–T groups and score meanings are project-defined. They have not been shown to form validated latent dimensions, to generalize across populations, or to measure anxiety, intelligence, diagnosis, or personality.

Independent observers are needed to establish whether the definitions can be applied reproducibly. Cohen's kappa measures chance-adjusted agreement, and weighted kappa can penalize ordinal differences differently [4, 5]. PSYCON preserves original ratings and adjudication separately so agreement can be evaluated before differences are resolved. High agreement would support reproducible use of the rubric in the studied setting, but would not establish every psychological interpretation attached to it.

The study also needs evidence that the recording and speaking opportunity support each score. A zero means the behavior was not observed despite a fair opportunity; N/O means opportunity or recording support was insufficient. Missing means no answer was supplied. Treating all three as zero would train on a changed target and could make poor recording access look like favorable behavior.

2.3 What the project adds and what it does not

Pretrained speech models provide transcription and diarization, including Whisper and Community-1 [6, 7]. They do not supply independent psychologist judgments of a participant's response to an exchange. PSYCON's contribution is the explicit connection between source evidence, reviewed human observations, annotation revisions, and independent evaluation units. The novelty claim is this project-specific measurement workflow, not that turn-taking or behavioral observation has never been studied.

The closest technical foundations answer different questions: Whisper transcribes speech; Community-1 separates anonymous speakers; TalkNet examines audiovisual speaking activity; and SpeechBrain provides speech-processing components [8, 9]. Their published results cannot be transferred to A–T prediction. For the behavioral study, they are possible sources of measurement error and tools whose outputs require review.

Data leakage is another methodological concern. Shared participants, repeated clips, and derived recordings can make a nominally held-out result partly familiar to a model [10]. The study therefore groups dependent people and sources before splitting examples. This design choice matters because a clip-level success can otherwise be mistaken for generalization to a new person.

3 Hypotheses, variables, and study design

3.1 Two separate behavioral evaluations

Hypothesis H1 is that supported speaker-specific measurements and recorded interaction context predict some independently reviewed A–T item ratings with lower held-out mean absolute error than a constant training-median prediction. The null comparison is no improvement over that baseline. The hypothesis does not require every item to be predictable or every score to be equally frequent.

Hypothesis H2 is that richer report representations receive better independently reviewed evidence-grounding and usefulness judgments than a transcript-only representation, without increasing unsupported psychological claims. The current application compares transcript-only A, structured-context B, and full-PSYCON C. B and C share frozen supervised predictions when available. The report study is distinct from prediction: a model may agree with a rating while producing an unhelpful report, or a report may be useful without a fitted predictor.

The implemented C representation adds history, retrieved evidence, and exploratory reference information together. It cannot isolate a history-specific effect. A future matched ablation must vary history while holding evidence retrieval and references fixed before H2 can be narrowed to a causal claim about history. The application currently supports a package comparison, and the paper reports that boundary.

Table 1: Behavioral variables and controls. Proposed outcomes are not measured results.

Study	Predictors or comparison	Outcome and controls
Rating prediction	Speaker timing, attributed words, marker proxies, context, supported event windows	Per-item reviewed 0–4 score; participant/source-held-out partitions, training-only preprocessing, fixed baseline.
Report comparison	A transcript; B structured context; C full representation	Blinded grounding, attribution, contextual appropriateness, usefulness, overclaiming; same speaker, transcript window, model digest, decoding settings.
Rater reliability	Independent ratings of the same source-supported behavior	Per-item raw and weighted agreement before adjudication; common instructions and opportunity definitions.
History-only follow-up	Earlier eligible history present versus withheld	Paired report review with retrieved evidence and reference information held fixed; not yet implemented as an isolated experiment.

3.2 Population, recruitment, and independence

The intended human reference comes from school psychologists reviewing group-discussion behavior. The retained engineering corpus contains eleven recordings and 85 marked person-recording places. Author-supplied seat correspondences establish repeated people across seven recordings. Nineteen links merge the 85 slots into 66 provisional identity components, an upper bound on unique people rather than a confirmed sample size because unlisted overlaps remain unknown. The original coverage denominator remains 85 slots. Recruitment procedures, demographics, and an approved final behavioral sample size are not established by the benchmark.

The supplied mapping corrects left to right only in its first section. It links six right-indexed seats in the afternoon 1.15.33 WhatsApp recording to IMG_3962; all eight same-position seats in the 10.19.44 and 10.19.02 recordings; one person across the 1.16.36 recording, IMG_3997, and IMG_3962; and three seats between 10.19.02 and IMG_3983. Directions are relative to the displayed frame, not a diarization index. The companion `docs/research/PARTICIPANT_OVERLAP.json` preserves the exact pairs without publishing names or photographs.

These links connect the eleven recordings into six provisional recording components. Four recordings form one component and three form another; four remain singleton recordings. For unseen-participant evaluation, the recordings within each linked component must stay in the same partition. Additional participant or source links may merge components further, so six does not establish six independent groups. Author-supplied correspondence improves provenance but does not validate every word-to-speaker assignment or supply consent. No live person mappings or training partitions were changed by this documentation update.

The prospective behavioral study needs a documented sampling frame, voluntary recruitment, inclusion/exclusion rules, language access, relevant accommodations, and reviewed repeated-person mappings. A school convenience sample can

support conclusions about the studied recordings, but cannot represent all adolescents or workplace communication. Moderator selection and topic assignment can also affect who gets a chance to display an item.

No formal power calculation is reported. The estimator's minimum data requirements are engineering gates, not a sample-size justification. Planning should use the desired precision of the primary paired outcome and the number of independent participant/source groups. Thousands of overlapping clips from a few discussions would not replace that independent sample size.

4 Operational measures and psychologist data

4.1 A–T observation definitions

The deployed form is derived from the tracked version 4.0 marksheet and interpretation guide. Its five areas appear in Table 2. Each label concerns one participant in one session. Observers retain confidence, fair opportunity, recording support, context, and timestamped evidence rather than contributing an undifferentiated personality total.

Table 2: Source-defined observation areas. These are operational groups, not validated psychological factors.

Area	Items	What is observed
Discussion tracking	A–D	Following the current point, addressing it, adapting to changes, and needing repetition.
Contribution structure	E–H	Idea sequence, requested support, completion, and task relevance.
Turn-sharing and reciprocity	I–L	Entry into turns, continued overlap, sharing cues, and recognition of a direct peer contribution.
Challenge-linked change	M–P	Observable change after disagreement, duration, participation, and feedback response.
Pressure-linked delivery	Q–T	Rate, hesitation, vocal, movement, or participation change against an earlier same-session baseline.

Scores range from 0 to 4. One means a weak or brief occurrence; two means repeated or noticeable behavior with limited effect; three means clear repeated behavior or an effect on the exchange; four means sustained behavior or a strong effect. The guide's interpretation thresholds are project rules, not scientifically calibrated cutoffs. N/O supports no conclusion about the absent observation, and missing answers remain missing.

A positive score requires contextual evidence. Item K and M–T need event and response windows; Q–T also need a baseline from earlier in the same discussion. For example, a reduced-participation label should include the challenge, earlier participation, later participation, and the opportunity to speak. Visible posture or gaze can be retained as human observation, but it cannot automatically train the audio-based predictor.

4.2 Scanned material and the form-version problem

Ten local scanned PDF bundles contain 228 image-only pages. The current import retains 76 participant forms from ten recordings as version 5.0, with A–T and P1–P12 role-related items. Across 2,432 answer records, 750 are numeric scores, 1,662 are N/O, and twenty are missing. The separately versioned source form is not silently renamed as the legacy version 4.0 rubric.

Imported transcriptions preserve their source version and answer states. Independent transcription review and rubric validation remain necessary; reviewer provenance is missing on 1,120 answer rows. Private scans, signatures, participant responses, and identifiable photographs are excluded from the paper and public figures.

Permission, ancestry, participant attribution, and independent label review remain separate gates. User-attested permission records are available, but item opportunity, observer confidence, structured evidence, validity checks, and independent reviews are absent. Forty-five imported mappings are stored as confirmed by historical model-supported seat transfer, seven are uncertain, and twenty-four are unmapped; stored confirmation is not independent human speaker validation. No answers meet strict behavioral eligibility. A separate source-score prototype retains those limitations explicitly.

4.3 Review, eligibility, and unavailable targets

The instrument stores observer answers and self-reports separately. It keeps original submissions, independent review, disagreements, adjudication, later corrections, and attachment provenance. Training requires documented permission, a current confirmed speaker mapping, source review, recording support, opportunity, observer confidence, and supported evidence. Reviewer repetitions are not additional independent examples.

Conflicting ratings require adjudication before supervised use. Original disagreement remains available for reliability analysis. Positive evidence must belong to the target speaker, event responses must follow the event, and same-session baselines must precede it. Missing and N/O answers are excluded from numeric fitting rather than imputed as favorable scores.

Item S remains excluded because supported vocal-change window predictors are absent from the current extractor. T may use supported participation change, while visual-only movement or gaze ratings remain stored and excluded from audio training. Independently defined communication targets have their own annotation requirements and are not generated from archetype preferences, A–T totals, or predicted answers.

5 Procedure and software-supported measurement

5.1 Recording-to-evidence procedure

The canonical instrument accepts supported audio/video files with original filename, recording time and timezone, context, anonymous participant codes, dataset role, and consent status. Originals retain their hashes and registered local paths. Exact hashes detect duplicate bytes, while reviewed source ancestry handles excerpts and re-encodings. Registration and queuing do not imply completed analysis.

The media worker records eight inspectable stages: preprocessing, diarization, transcription, attribution, segmentation, features, evidence, and profile. Audio is normalized to mono 16 kHz PCM with a playback copy separate from the original. Duration, clipping, silence, and corruption diagnostics can withhold unsupported analysis. An energy-contrast value is an SNR proxy, not a calibrated measure of recording noise. Aggressive denoising is disabled.

Community-1 creates anonymous intervals, and faster-whisper large-v3 provides timestamped transcription [11]. The present transcription adapter fixes English, voice-activity filtering, beam size five, and disabled conditioning on previous text. These settings do not establish multilingual validity. The default limits are twelve speakers, four hours, and an 8 GiB upload per recording.

Word attribution requires strong interval coverage for one speaker and little competing coverage. Ambiguous overlapping words remain unattributed. Turn/utterance segmentation and neighboring exchanges create the behavioral context. A reviewer then connects a session participant to the correct anonymous cluster. Persistent person identity, session participant identity, and cluster identity remain distinct.

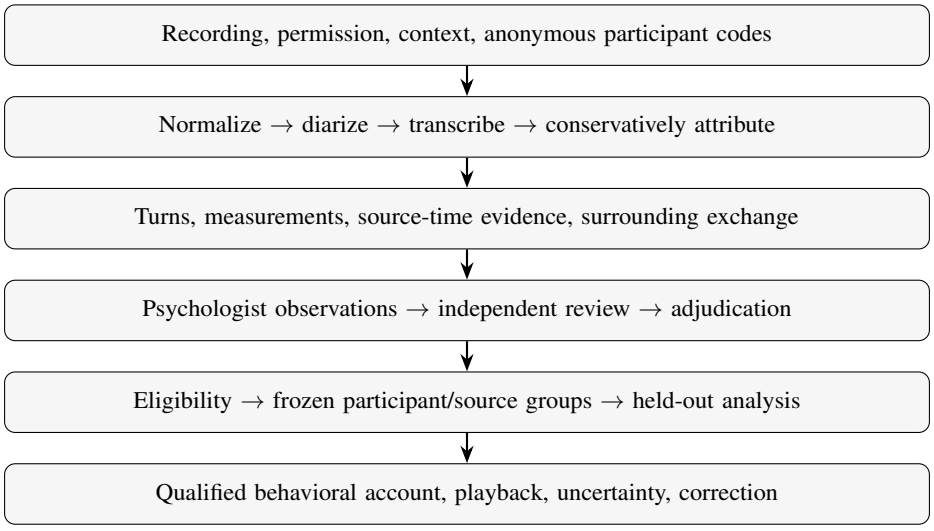


Figure 1: Study procedure supported by the implemented instrument. Human observations enter through independent review. Diagram generated with AI-assisted LaTeX/TikZ from the reviewed code and form definitions.

5.2 Conversational measurements and denominators

Speaking time is the union of a speaker's diarized intervals, including overlap. Speaking share uses exclusively attributed non-overlapping duration, $q_i = D_i^{\text{exclusive}} / \sum_j D_j^{\text{exclusive}}$. Turn count follows merging same-speaker

intervals separated by at most 0.3 seconds. Words per minute divides attributed words by diarized speaking duration; incomplete attribution can lower this value, so it is not a complete measure of fluency.

Overlap-entry candidates require at least 0.5 seconds of entry and at least 0.2 seconds remaining in another speaker's turn. Their rate uses session duration and is expressed per ten minutes. This detects a temporal relation; interruption intent remains a human interpretation. Directed-response estimates use adjacent turns and bounded gaps, which can misidentify the intended addressee in a group.

Transparent English rules count question punctuation and selected proposal, hedging, acknowledgement, disagreement, reasoning, experimentation, and concession markers. Ratios use attributed utterances, while rates use session exposure. These are low-confidence proxies. Saying "because" does not establish reasoning quality, and punctuation does not independently determine the function of a question.

Manual event annotations can supply richer topic, argumentation, question-type, and turn-sharing descriptions. Their coverage is explicitly partial. Automatic topic control, persuasion, paraphrasing, and interruption intent remain unvalidated or unavailable. A missing semantic measure is not treated as proof that the behavior did not occur.

5.3 Event change and personal history

The Q–T baseline occurs before an identifiable event in the same session. Historical comparison instead asks how the current session differs from earlier conversations. Historical summaries use strictly earlier eligible completed recordings linked to a reviewed person, exclude reference-only material and related source duplicates, and restrict use by dataset role. One long recording does not gain extra weight simply because it contains more windows.

The implementation retains recording-level means, medians, variance, quantiles, sample counts, and source lineage. Availability requires five comparable prior recordings. A difference exceeding two sample standard deviations is a descriptive prototype flag, not a significance test, and five recordings do not establish stable personality. Neither same-session change nor historical deviation shows that a challenge caused the observed response.

6 Analysis plan and implemented predictor

6.1 Independent units and frozen datasets

The snapshot builder retains labels, mappings, source hashes, consent and review records, feature versions, and partition assignments. The example unit is a participant in a session, with separate supported item targets. Connected components join recordings through confirmed repeated people and shared/reviewed original sources, and cannot cross unseen-participant train/validation/test boundaries. Reference overlap and incompatible roles are excluded.

This grouping may connect much of a small corpus, leaving too few independent groups to fit. Randomly splitting clips would hide that limitation rather than solve it. A separate longitudinal protocol permits repeat-person evaluation with explicit training/validation/evaluation roles and strict forward-time cutoffs. Unseen-person generalization and within-person prediction are different estimands and require separate reports.

The current supervised snapshot uses conversational measurements, context indicators, transcript confidence, and supported target-specific event-window features. Historical report summaries are not automatically inserted as predictor inputs. A prediction claim about personal history therefore needs an additional explicit feature experiment. This separation prevents a feature that only exists in reporting from being described as trained supervision.

6.2 Estimator, baseline, and selection

Supported targets use median imputation with missingness indicators, scaling, and class-balanced logistic regression from scikit-learn [12]. The estimator approximates ordinal 0–4 scores as categorical classes; it does not enforce an ordinal threshold structure. Default regularization candidates are $C \in \{0.1, 1, 10\}$, selected by validation mean absolute error. Preprocessing fits on training data, and final test examples do not select the configuration.

At least twelve training examples, three independent training groups, two score classes, and supported measured predictors are required. These minimums are software gates, not evidence of statistical power or adequate population coverage. If validation is absent, the implementation avoids hyperparameter search and marks unsupported evaluation as exploratory.

The constant comparator predicts the integer training median. The portable artifact retains feature order, medians, missingness indicators, scaling, classes, coefficients, and intercepts, and its predictions are checked against the fitted

estimator. A model remains a candidate until deliberately activated. Predictions retain model/input lineage, and correction or withdrawal identifies dependent artifacts requiring invalidation or retirement.

6.3 Statistical outcomes and uncertainty

For each target, the primary prediction outcome is $\mathrm{MAE} = n^{-1} \sum_r |y_r - \widehat{y}_r|$. The code also records macro-F1, balanced accuracy, a five-class confusion matrix, quadratic-weighted kappa, and multiclass Brier score. Probabilities are uncalibrated and cannot be presented as validated confidence. A 500-draw connected-group bootstrap supplies an MAE interval when enough independent held-out groups exist.

The software's "evaluated" gate requires minimum validation/test counts and MAE below the constant baseline in both partitions. This status is not a statistical significance test, a validity certification, or a deployment approval. Coverage computed on retained examples is not coverage of all stored sheets. Final analysis must report exclusions, N/O, missingness, abstention, class distributions, participant counts, independent-source counts, and uncertainty.

The prospective study should predeclare one primary target or aggregate outcome justified by the behavioral question, its analysis unit, and the meaningful improvement. Per-item secondary comparisons need a multiplicity policy. Rater reliability should be assessed before adjudication, and small or constant-score samples should be reported without forcing a kappa estimate. No p-values or confidence intervals for human outcomes are invented in this paper.

6.4 Blinded report comparison

A uses the first forty selected utterances around the target speaker's first contribution, including neighboring speakers for context. B adds session measurements, interaction context, and frozen predictions when available. C adds earlier history, retrieved speaker evidence, and exploratory references. The same transcript window is used in all three conditions; it is not a full-recording interpretation.

The local Ollama model's actual digest, prompts, inputs, output, attempts, and decoding settings are stored. Defaults use temperature zero, seed 42, a 16,384-token context setting, and 3,200 generated tokens. At most 24 retrieved evidence records are selected across indicator types. Invalid, wrong-speaker, oversized, and truncated outputs are withheld rather than substituted with invented claims.

Evidence-ID validation checks reference integrity, not semantic support. Independent reviewers must play the cited source and judge whether the claim follows. Planned primary review dimensions are grounding, correct attribution, contextual appropriateness, usefulness, and overclaiming. Exploratory role-reference agreement is secondary. Randomized condition order, masked system labels, common source material, and retained failed runs are necessary controls; no completed blinded behavioral outcome is reported.

7 Current results and evidential boundaries

7.1 Behavioral outcomes remain pending

The release does not supply independently validated held-out A–T prediction results, psychologist inter-rater agreement, blinded usefulness scores, or longitudinal improvement estimates. Imported raw scores support an exploratory computational fit, not those outcomes. H1 and H2 remain untested in the reported human data.

This boundary is the main behavioral finding of the current evidence audit: the instrument supports a testable procedure, but the available artifacts do not yet answer the behavioral question. A software demonstration is not an effect size about people. The evidence does support reporting recorded engineering comparisons and explaining the remaining measurement gates.

7.2 Retained profile-coverage comparison

The earlier audiovisual comparison used eleven videos and 85 marked person-recording slots. It combined anonymous turns, TalkNet visual observations, and SpeechBrain voice checks; Nemotron supplied additional diarization paths [13]. Table 3 reports the committed aggregate artifacts.

Guarded PSYCON retained 57 profiles, a descriptive 67.1% of slots, compared with 51 or 60.0% for each guarded Nemotron path. The difference is six slots, or 7.1 percentage points of coverage. No participant-level uncertainty or statistical significance is claimed because this is a saved corpus comparison without independent identity ground truth. The denominator is not unique people.

Table 3: Engineering context only: ready-profile counts and usable duration. These are not behavioral validity or identity-accuracy scores.

Method	Ready profiles / 85	Usable seconds
Original pipeline	40	395.14
Earlier NVIDIA pipeline	13	524.92
First PSYCON	45	4,364.61
Guarded PSYCON	57	4,640.04
Guarded Nemotron streaming	51	4,849.58
Guarded Nemotron offline	51	5,179.03

The gate required at least three usable clean seconds and a valid voice embedding. Guarded PSYCON improved the first PSYCON count from 45 to 57. Nemotron offline recovered four slots that PSYCON left incomplete and lost ten that PSYCON retained; five intervals disagreed with retained PSYCON visual observations. Either detector can be wrong. More audio seconds did not necessarily mean more covered slots.

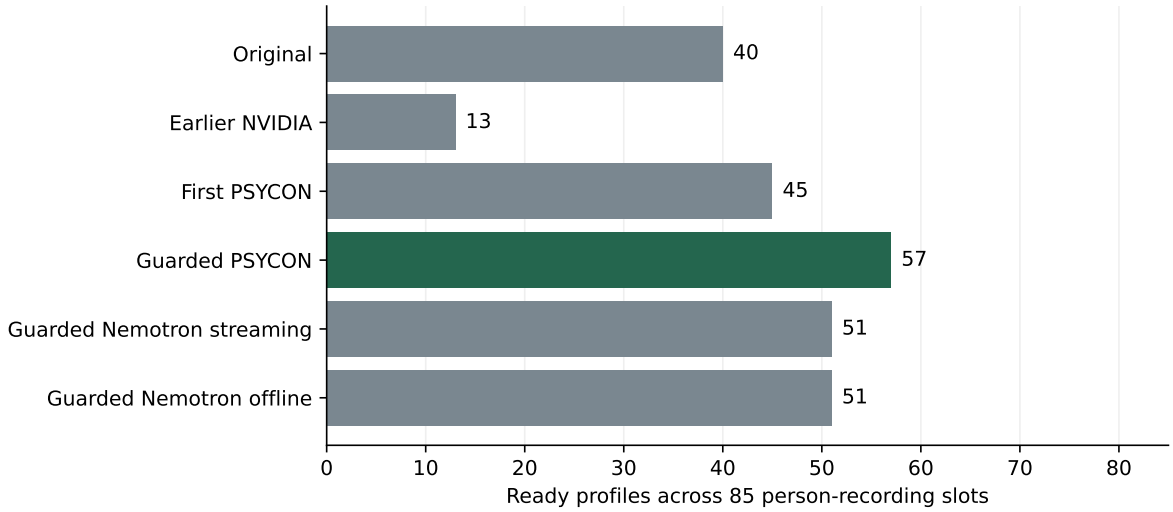


Figure 2: Aggregate coverage regenerated from the committed benchmark JSON using AI-assisted Python/matplotlib code in the companion builder. This describes the earlier audiovisual workflow, not psychological accuracy, and contains no participant images.

The earlier NVIDIA path had eighty anonymous channels but thirteen ready face-linked profiles. Channel count and ready-person-recording profile count are different measurements. Threshold diagnostics and saved detector disagreements are review aids, not independent labels. This benchmark does not establish the active audio-only instrument's attribution accuracy or any human behavioral hypothesis.

7.3 Completed exploratory fit on real source scores

The author-supplied overlap links were applied to 34 participant records representing fifteen known people. Eight linked seats were registered without answers or permissions, bringing the live inventory to 84 participant-session records; 76 have imported forms. Application retained private before-state journals and database audit events. These links constrain grouping, but do not validate word attribution or resolve every repeated person.

An explicitly separate source-score task retained 488 numeric observations across forty participant-session examples, nine recordings, and five known person/source groups. It excluded N/O, missing answers, uncertain or absent speaker mappings, unavailable permission, invalid source ancestry, conflicting ratings without adjudication, and unsupported S/T targets. It retained absent opportunity, confidence, evidence, and independent-review fields as absent, with unmet validation requirements attached. No synthetic participant observation replaced a missing answer.

Fourteen targets passed the engineering fitting gates of twelve examples, three groups, and two score classes. Their models used 423 labels in total, median imputation with missingness indicators, standard scaling, and class-balanced

logistic regression at fixed $C = 1.0$, seed 42, and maximum 2,000 iterations. All retained examples were assigned to training: there was no regularization search or independent validation/test set. P1–P12 source role ratings used general conversational proxies and have no demonstrated occupational or psychological validity.

Table 4: Real source-score labels used by the exploratory Docker fit. Counts are item observations, not unique people or measured accuracy.

Target	Labels	Groups	Target	Labels	Groups
G	13	5	I	13	4
P1	34	5	P2	35	5
P3	33	5	P4	37	5
P5	29	5	P6	27	5
P7	36	5	P8	33	5
P9	35	5	P10	35	5
P11	30	5	P12	33	5

The Docker run completed, produced fourteen portable model artifacts, and activated them with exploratory acknowledgement. An API check returned fourteen scored predictions marked exploratory. Eighteen source targets remain untrained: fifteen retained targets were too sparse, L had no retained scores, and S/T lacked supported predictors. The aggregate release records snapshot hash, runtime, target counts, and limitations; weights remain private. Execution and successful prediction do not establish generalization, calibration, rater reliability, or either behavioral hypothesis.

7.4 Retrospective omitted-group diagnostics

After the final fit, a separate exploratory analysis left out one of five known person/source groups at a time and refitted the same fixed-C estimator on the others. Feature availability, imputation, scaling, and median baselines were derived from each fold's training data only. Original fitting gates were reapplied; inadequate folds abstained. This retrospective analysis was not preregistered and cannot establish an independent prospective final-test result. Source ratings remain unreviewed, and unresolved overlaps may still permit leakage.

The deployed artifacts reproduced 414 of 423 training labels exactly (97.9%), with mean absolute error 0.026 score units. Omitted-group models scored 402 labels, abstained on twenty-one, and matched 230 exactly (57.2%). On the same scored labels, fold training-median baselines achieved 64.4% agreement. Model MAE was 0.562 versus baseline MAE 0.460. Thus the pooled raw-score model underperformed this simple baseline. P1–P12 supplied 397 scored observations; G and I supplied only three and two. This chiefly measures role/task source-score approximation, not comprehensive A–T generalization. The gap is consistent with overfitting or group differences, but does not identify their cause or formally settle H1.

7.5 Retrospective model repair and class sensitivity

A subsequent development analysis compared nine disclosed candidates: training median and mode references, logistic classifiers at $C = 0.01, 0.1, 1$ without class balancing and $C = 0.01, 0.1$ with balancing, and rounded ridge predictions at $\alpha = 10, 100$. Each outer fold omitted one known connected group. Selection used only inner omitted-group folds from the remaining groups, minimizing equal-group MAE within each target; preprocessing and feature availability were derived from the relevant training fold. A candidate needed an inner MAE improvement of at least 0.05 over the median reference, otherwise the reference was retained. The original count/group/class fitting gates applied to feature-based fits; inadequate fits used an explicitly labelled training-median reference. This policy was developed after inspecting the earlier results, so nested splitting does not make it preregistered or an untouched independent test.

On the same 402 item observations evaluated previously, nested selection achieved 60.9% exact agreement and MAE 0.493, compared with original 57.2%/0.562 and median-reference 64.4%/0.460. The 3.7-point accuracy gain and approximately 12.4% error reduction did not establish superiority to baseline. Mean sensitivity over observed score classes decreased from 30.3% to 28.8%; median-reference sensitivity was 29.6%. Sensitivity is exact-score recall, not clinical sensitivity or psychological validity. Original recalls for scores 1–4 were 15.4%, 0%, 33.0%, and 72.9%; revised recalls were 15.4%, 0%, 13.6%, and 86.1%, on respective supports 13, 20, 103, and 266. Score 0 had no observations, so its sensitivity is unavailable. Pooled targets have different behavioral meanings and do not constitute one latent construct.

Including reference fallbacks, the revised procedure covered all 423 labels with 60.0% agreement and MAE 0.508, versus median-reference 63.4%/0.478. A paired bootstrap resampling five known groups produced a descriptive 95% percentile interval of 0.009–0.052 score units for model-minus-reference MAE; few groups, unresolved overlaps and

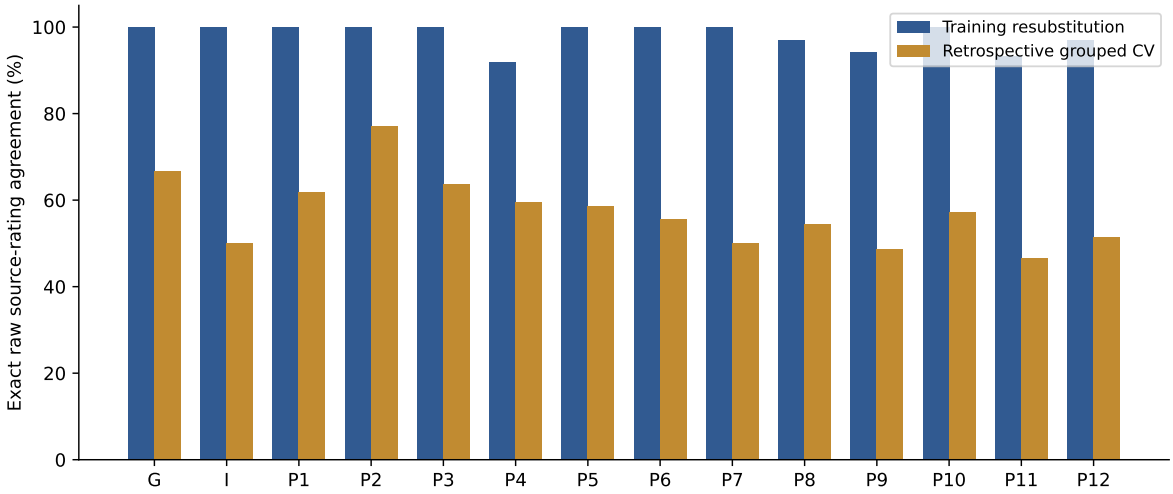


Figure 3: Exact raw source-score agreement by target, generated from the committed aggregate cross-validation summary. Training resubstitution uses deployed artifacts; omitted-group diagnostics refit separate models. G/I CV denominators are three/two; the P targets have 27–37 observations each. These are repeated item labels, not independent people, and no validated psychological accuracy is claimed.

retrospective development prevent a population or significance claim. Final fitting selected nine median references and five feature-based estimators. Fourteen portable fits were registered as exploratory candidates and replayed against all 423 training labels, with 77.3% training agreement. They were not promoted because the evaluated selection procedure remained inferior to baseline. All attempted results, class supports, precision, specificity and fold choices remain inspectable.

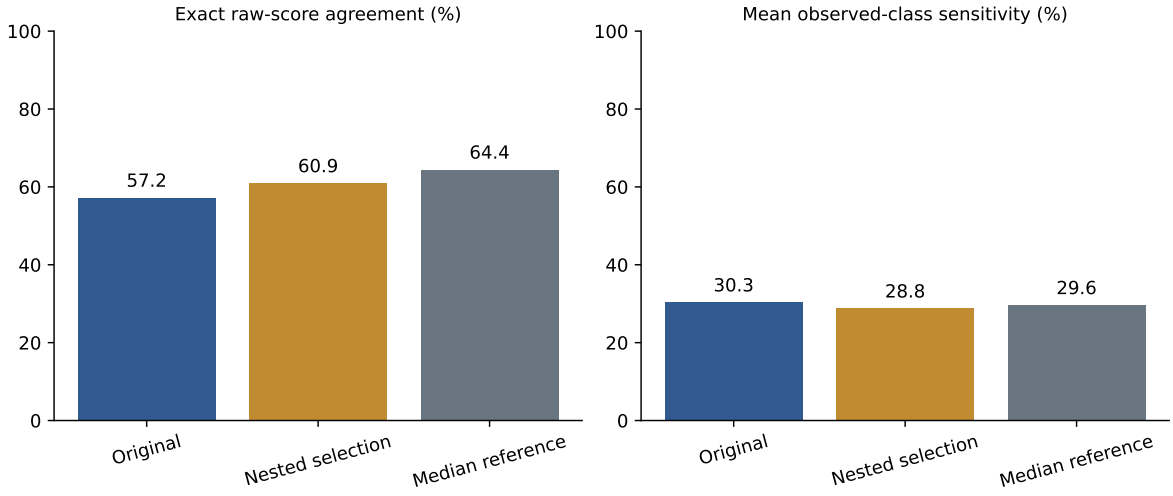


Figure 4: Retrospective repair on the identical 402 item observations. Higher exact agreement came with lower class-average sensitivity; both feature-based procedures remained below the median reference in exact agreement. AI-assisted Python/matplotlib regenerated these aggregate figures. There is no independent final-test result.

A separately disclosed post-hoc sensitivity-oriented experiment used the same nine candidates, selecting within inner folds by pooled observed-class macro sensitivity and requiring a 0.02 gain over the median reference. On the identical 402 observations it achieved 56.5% agreement, MAE 0.562 and 29.9% mean class sensitivity. Score-specific recalls for 1–4 were 15.4%, 5.0%, 24.3% and 74.8%. Detecting one of twenty score-2 observations did not resolve minority-score failure; class-average sensitivity remained below the original 30.3%. All 423 labels including reference fallbacks yielded 55.8% agreement and MAE 0.574. This additional attempt was not promoted or presented as an independent confirmation. Changing the selection objective did not establish reliable individualized prediction.

7.6 Compact predictors and ordinal modeling

A further disclosed development procedure addressed the mismatch between 31 available predictors and forty participant-session examples. Four predefined core measurements were speaking share, words per minute, mean turn duration and response latency. A second set added question rate, proposal rate, acknowledgement ratio and interruption-candidate rate. These remain automated measurements or rule-based proxies, not validated psychological constructs. Availability was determined only within the relevant training fold, with no identity, topic or other-target response features added.

Twelve fixed candidates comprised median/mode references, core logistic models at $C = 0.1, 1$, compact logistic models at $C = 0.1$ with and without class balancing, core/compact ridge at $\alpha = 10$, compact cumulative-threshold logistic models at $C = 0.1, 1$, and compact random-forest/extra-tree regressors. Ensembles used 64 trees, depth at most three and at least four examples per leaf. Ordinal fits independently estimated $P(Y > k)$ for $k = 0, 1, 2, 3$; cumulative-minimum projection enforced nonincreasing exceedance probabilities, followed by a lower-median ordinal decision. These probabilities are uncalibrated. Regression produced rounded bounded scores without invented class probabilities. Logistic ties within 10^{-12} probability units preferred the lower class, while older artifacts retained their previous tie behavior.

Inner selection minimized equal-group MAE among candidates whose pooled observed-class sensitivity was no more than 0.02 below the median reference, and required a 0.025 MAE gain over that reference. Otherwise it retained the reference. The twelve-example/three-known-group/two-class fitting gates and training-only preprocessing remained in force. This sensitivity constraint governs inner selection; it cannot guarantee recall in an omitted outer group. The policy was fixed before this particular run, after previous results had been examined, and therefore is retrospective rather than preregistered.

The compact procedure achieved 250/402 exact matches (62.2%), MAE 0.490 and mean class sensitivity 30.0% on the common comparison labels. This is a 5.0-percentage-point agreement increase and approximately 12.8% lower error relative to the original model, but it remains worse than median-reference 64.4%/0.460 in agreement/error. Exact-score recalls for scores 1–4 were 15.4%, 0%, 18.4% and 86.1%, with supports 13, 20, 103 and 266. No reliable minority-score detection or broad behavioral validity follows.

Across all 423 labels, including reference fallbacks, compact selection produced 61.2% agreement and MAE 0.506 versus median-reference 63.4%/0.478. A paired five-known-group bootstrap yielded a descriptive model-minus-reference MAE interval of $[-0.013, 0.049]$, spanning zero. Few groups, unresolved overlap and prior examination prevent a superiority or population claim. Final fitting produced eight population references and six feature-based models, with training replay agreement 73.5%. Fourteen artifacts were saved as exploratory candidates without promotion. All twelve fixed-candidate outer diagnostics were retained; choosing their apparent winner would add selection bias. For example, mode-reference agreement was 64.5% over all 423 labels, further emphasizing the need for multiple simple controls.

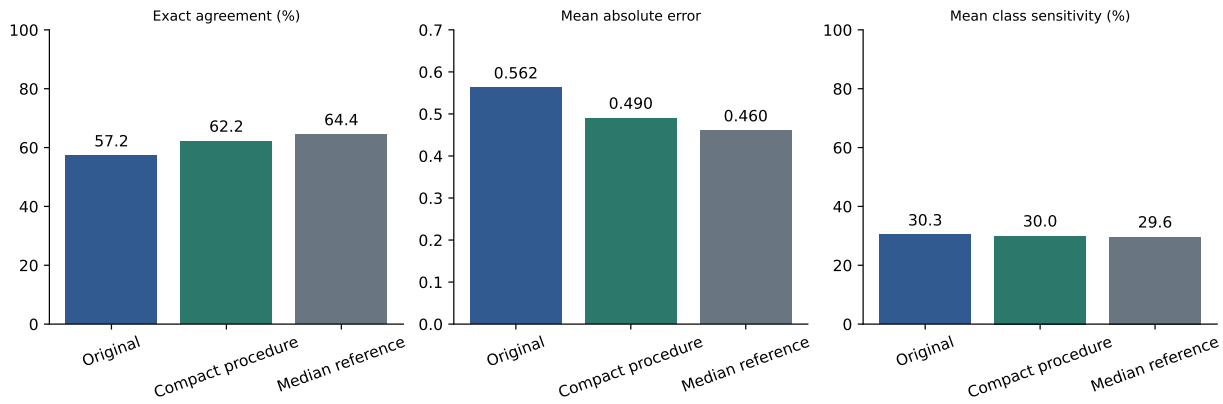


Figure 5: Earlier compact procedure compared on identical 402 source-score observations. Reduced complexity improved agreement and ordinal error relative to the original model, but the median reference remained better on those measures. Average class recall remained low. No independent final-test outcome is available.

7.7 Maximum defensible performance audit

On 13 October, a separate frozen-input audit checked answer IDs, target/speaker alignment, ordinal scales, duplicate records, missing outputs, denominators, source hashes and known person/source transitivity. Refitting the original, compact and previous agreement-selection procedures reproduced every saved prediction. No demonstrable implementation defect explains their agreement. The 423 fitted-target ratings have score counts $(0, 24, 24, 109, 266)$ for categories 0–4. The common 402-rating cohort has counts $(0, 13, 20, 103, 266)$; it is fixed by original prediction availability, not chosen for favourable outcomes.

A bounded second grid used penalized multinomial and cumulative ordinal logistic models with core, conversation and extended existing-rate features. Professional targets P1–P12 shared coefficients with penalized target indicators; G/I were fitted separately because their score direction differs. The grid included $C = 0.01, 0.1, 1$ for core/conversation logistic models, a training-balanced conversation model, an extended-rate model and conversation ordinal models at $C = 0.1, 1$, with median/mode references. Each outer fold omitted a known connected group. Training-only imputation and scaling preceded inner grouped selection; recording/person identity, observer notes, annotation windows and transcript confidence were excluded. These are completed-session inputs, not evidence of real-time prediction or calibrated probabilities.

Inner selection maximized equal-group agreement subject to no MAE increase over either reference, individual-class recall loss at most 0.02 versus median for classes with at least five inner observations, and a minimum 0.025 agreement gain over the best reference. The disclosed policy was frozen before these runs after earlier results were known, so it remains retrospective development. Promotion required agreement gain over compact, no MAE increase versus compact/mode, no score-specific recall loss above 0.02 versus compact, and positive descriptive paired agreement lower bounds versus compact/mode. Fixed outer diagnostics could not select another winner.

Table 5: Verified common-cohort comparison. All rows score the same 402 ratings; the original abstains on 21 of 423. Other procedures score all 423, often with explicit reference fallbacks.

Procedure	Correct	Agreement	MAE
Original grouped model	230	57.21%	0.562
Compact procedure	250	62.19%	0.490
Previous agreement selection	247	61.44%	0.498
Fold training median	259	64.43%	0.460
Fold training mode	264	65.67%	0.453
New nested pooled procedure	264	65.67%	0.453

The new procedure chose references in four folds and core logistic $C = 0.01$ in the fifth, producing exactly the same predictions as the mode reference. Score 1–4 recalls were 15.38%, 0%, 4.85% and 96.62%, versus compact's 15.38%, 0%, 18.45% and 86.09%. Balanced accuracy over observed truth classes was 29.21% versus 29.98%; quadratic weighted kappa was 0.129 versus 0.101. MAE treats adjacent score intervals as equal and kappa uses squared ordinal distances. Score 0 has no support, and pooled metrics mix distinct target meanings. Full precision, recall, F1 and confusion matrices accompany the aggregate release.

The strongest fixed diagnostic in that experiment achieved 270/402 agreement (67.16%), MAE 0.435, but predicted score 4 on 397 ratings and recalled only 1.94% of score 3. It is not the selected procedure; choosing it after observing outer results would add selection bias. The new procedure failed the rare-class and mode-superiority promotion gates, so compact remains the research reference and no model was activated. No result supports 85% agreement. On all 423 ratings, compact achieved 259 matches (61.23%, MAE 0.506), previous selection 256 (60.52%, 0.513), median 268 (63.36%, 0.478), and mode/new selection 273 (64.54%, 0.470), with no abstentions in these procedures.

All nineteen documented identity links and registered source hashes respect frozen components, but twenty-one participant-session identities remain unresolved. The fitted ratings lack independent reviews, recorded opportunity decisions, item confidence and structured evidence windows. This prevents an inter-rater reliability estimate or defensible numerical ceiling. Independent blinded re-rating must preserve originals; new identity-resolved connected groups are needed for final evaluation. Six audit regression tests contributed to 82 passing research tests, with one ffmpeg-dependent skip. The public aggregate release preserves every comparison without publishing individual IDs or notes.

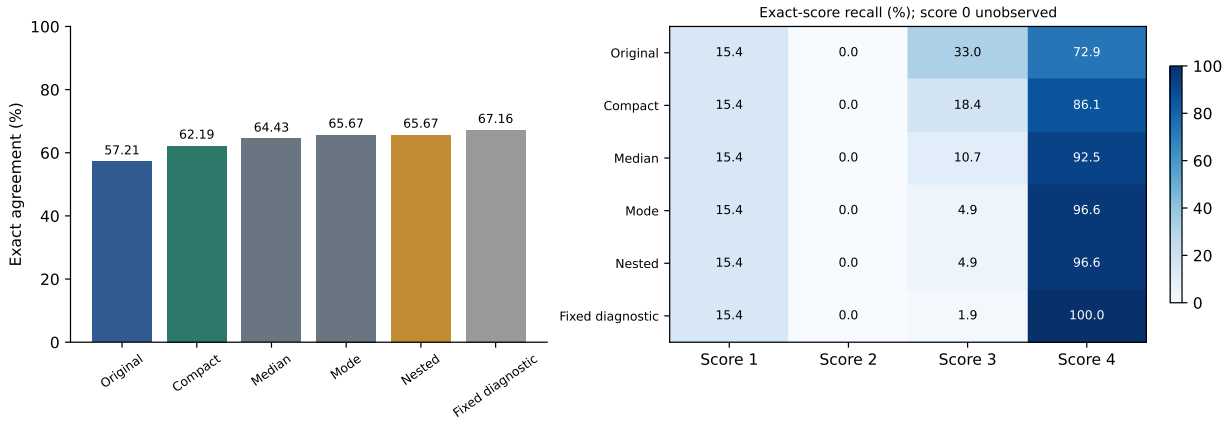


Figure 6: Verified agreement and class recall on the fixed 402-rating cohort. The nested challenger matches the mode reference; the higher fixed diagnostic largely predicts score 4 and is not promoted. These are retrospective development results, not independent final testing.

7.8 Full pipeline autopsy and feature-family ablation

A later bounded autopsy traced normalization, diarization, word timestamps, participant attribution, segmentation, observable measurements, training-only preprocessing, per-target prediction and reference alignment. All nine retained normalized recordings matched source audio duration within 15 milliseconds and remained below the clipping gate. Independently corrected transcripts and speaker timelines were unavailable, so no word-error, diarization-error or causal stage-error estimate was inferred.

Fifty-one declared configurations compared five feature families independently, the full 31 predictors, compact/core subsets, each family removed, training-only missingness/redundancy cleanup, log transformation and quantile clipping. Regularized logistic and ridge alternatives used the same five outer connected groups and inner grouped selection. The selected challenger achieved 251/402 agreement (62.44%), MAE 0.488, balanced accuracy 29.93% and quadratic weighted kappa 0.097, versus compact's 250/402, 0.490, 29.98% and 0.101. On all 423 source ratings it achieved 260 matches (61.47%, MAE 0.504), with no abstentions. It failed replacement gates and remains exploratory. The highest fixed diagnostic, timing/overlap logistic $C = 0.1$, reached 273/402 (67.91%, MAE 0.425); it was not selected by observing outer results or promoted.

A demonstrated extraction bug interpreted unordered retry rows as chronological turns, changing valid one-second response gaps to missing values. Schema 1.0.2 sorts turns before temporal measurements; a regression test covers every permutation of the three-turn example. Existing cached features, source ratings and saved model artifacts remain unchanged, so no historical accuracy gain is attributed to this fix. Focused validation passed 101 tests with one skip; PostgreSQL integration fixtures were unavailable. No family was deleted merely for low standalone performance. The compact eight-feature research reference is retained, and uncertain incremental contributions remain inconclusive. Aggregate ablations, all-target/class/group slices and component decisions are retained in `docs/research/pipeline-autopsy-2026-10-13/`; identity-bearing predictions remain private.

7.9 Current full-input report example

An earlier condition-C run used fourteen source-score predictions, contextual measurements, evidence and the selected communication goal with local qwen3.5:4b. Repeated model lineage, uncertainty, evidence lists and transcript exchanges were encoded through lossless references to satisfy the explicit packet budget. The example had zero eligible earlier recordings, so historical personalization was unavailable. The report passed schema and cited-ID integrity checks, but returned four claims with only two unique claims, mostly paraphrased the topic, and did not visibly use the ratings for communication coaching. Structural validation therefore does not establish semantic support or usefulness. A single generated report cannot answer H2; blinded, history-controlled human evaluation remains pending.

The revised reporting path separately supplies deterministic recording quantities, their meanings and an optional prewritten exercise selected by the user's goal. It marks the exercise as untested and does not infer it from source-score predictions. Revised validators withhold duplicate observations/inferences, incorrect measured values and unverified derived numerical inferences. Prompt version 1.3.3 further constrains the observation, feature reference, summary,

inference alternatives and selected-goal exercise. A real local run completed with speaking share 17.80%, a low-confidence contextual interpretation and the Executive exercise to state a decision, support it with a reason, and invite a response. The interface distinguishes full-recording measurements from linked transcript excerpts, which do not independently verify aggregate quantities. These bounds improve factual consistency by limiting generation; they do not demonstrate freely generated personalized coaching. Earlier failed attempts remain retained, and no eligible history or blinded benefit outcome was created.

A history-only input-pair helper holds all other representation fields fixed and refuses unavailable history. This repairs the comparison design but does not constitute a completed experiment. Feature-definition version 1.0.1 also corrects denominator metadata for turns, speaking duration, exclusive speaking time, question/proposal opportunities and response gaps. Numerical feature values and frozen source observations were not changed; earlier denominator metadata remains historical.

7.10 Workflow and synthetic demonstrations

Implementation notes record real-audio CUDA smoke processing, matched local A/B/C generation, and staged PostgreSQL/frontend checks. An isolated training demonstration used eighteen explicitly synthetic participant/session examples with twelve/three/three independent-group partitions. It exercised label intake, review, snapshot creation, fitting, activation, prediction, rollback, invalidation, and export.

Those results establish execution and lineage handling, not participant performance. Synthetic labels do not populate the canonical research result as human outcomes. Historical test counts apply to their recorded source revisions and environments. The build guide records the present document build separately and does not substitute old test counts for a fresh run.

8 Interpretation, alternative explanations, and limitations

The main threat is interpreting a conversational proxy as a psychological construct. Low word count can result from moderation or limited access; overlap can be normal cooperation; a marker such as “I agree” need not be sincere acknowledgement. The study must evaluate the actual operational item against source evidence instead of assuming that a plausible feature name measures the intended behavior.

The strongest alternative explanation for future predictor agreement is shared context or observer practice. A model could learn which topic, group, moderator, or recording quality tends to attract certain scores instead of learning a participant’s behavior. Group/source separation, context-stratified errors, independent observers, and feature ablations can expose this. A null or reversed improvement would challenge H1 and should be reported directly.

Another alternative is common-source error. Observers and the model could both rely on an incorrect transcript or cluster mapping. Reviewers should have access to original audio, and attribution error must be measured independently rather than inferred from rating agreement. Removing uncertain examples can improve conditional performance while reducing real-world coverage, so exclusion and abstention require equal visibility.

For reports, C may look better because it has more evidence or more polished information, rather than because earlier history is informative. The history-only ablation is the strongest single follow-up to isolate that interpretation. A report that receives higher usefulness ratings but more unsupported personal claims would not satisfy the intended benefit. Familiarity with a participant or inferred condition identity could also bias reviewers.

Human data remain limited by unresolved source-version differences, absent confirmed recruitment/sample-size information, and pending independent evaluation. A school convenience sample, fixed English processing, limited language access, unequal opportunities, and unvalidated semantic rules limit generalization. Current ordinal approximation and rare score classes further constrain estimator interpretation. Stable personality and causal effects are outside the evidence.

Practical problems included overlapping speech, short clean turns, ambiguous identity, slow remote-database reads, GPU contention, and preserving corrections during retries. Conservative attribution, durable jobs, adapter boundaries, grouped queries, and immutable revisions address operation, but do not remove psychological measurement limitations. Local single-user execution and incomplete physical-phone/media-backup verification also limit product readiness.

9 Future studies and potential impact

The next behavioral study should review the versioned 5.0 transcriptions and rubric crosswalk, verify permission and mapping independently, collect source-supported opportunity/confidence and independent ratings, and freeze an adequately sized participant/source split. It should report reproducibility of human scoring before model performance. If an item lacks agreement or adequate opportunity, the appropriate response is to revise or exclude it transparently.

The most informative report follow-up is a paired history-only experiment with the same speaker, transcript, evidence retrieval, reference information, local model, and decoding settings. Independent raters would judge support, usefulness, and overclaiming from source playback. The comparison can challenge the claim that personal history helps; collecting a larger sample without isolating these inputs would not resolve that mechanism.

Longer-term work could add reviewed semantic annotations, supported vocal-change windows, ordinal estimators, and calibration, with new feature versions and held-out comparison. The envisioned product is voluntary communication review that lets a person inspect and correct evidence and practice a specific adjustment. Demonstrated benefit in the studied population must precede wider school or workplace claims. Diagnosis, recruitment ranking, and disciplinary scoring are not supported uses.

Hardware remains a secondary future branch. Retained designs describe ESP32 wrist/audio modules, physiological sensors, an I2S microphone, timestamps, and BLE. Ear firmware contains I2S capture and protocol transport; wrist sensor packets still use placeholders. No tracked KiCad schematic, PCB, or validated manufactured assembly exists. A future three-dimensional assembly diagram, KiCad circuit/board, verified power protection, pin assignments, bill of materials, and bench records would prepare a separate approved synchronized-sensor study. They do not supply current behavioral results.

10 Ethics, availability, and student contribution

Recorded discussions and handwritten ratings can identify participants even when names are replaced with codes. New recruitment, behavioral data collection, and applicable reuse require the relevant prior review and consent/assent processes. Society for Science human-participant rules distinguish approved studies and specific exemptions [14]; this paper does not assert that the project's existing recordings qualify for an exemption. On 12 October 2026, nine privately supplied photographs of signed group-session participation sheets were visually reviewed. Their printed wording establishes agreement to a voluntary discussion, the right to stop, and no medical or psychological diagnosis. Some session fields are incomplete and people repeat across sheets, so nine sheets is not a unique-participant count or verified recording coverage. The authors confirm that the complete records document recording and model-training permissions, applicable parental permission and prior approval, including details outside the supplied photographs. This is an author-confirmed account; those portions and consent timing have not been independently checked. No approving-body name, approval identifier or approval date is inferred from the images.

Training permission is separate from retaining a recording. The instrument records permission, exclusion, correction, and withdrawal and invalidates dependent lineage. Existing weights may require retirement or retraining; this is not a claim of exact machine unlearning. Confidential source access, volunteer participation, psychological/social risk review, and prevention of evaluative misuse belong in the approved study protocol.

The repository contains implementation, rubric sources, synthetic tests, aggregate benchmark data, and archived physiological metrics [15]. Private original recordings and the local scanned bundles are not published here. A future reproducible behavioral evaluation needs a governed access process or permitted deidentified package, reviewer instructions, frozen manifests, and an explicit source/version record.

The prior manuscript attributes software and machine-learning work to Arjun Vijay Prakash and hardware/system work to Saksham Yadav. That attribution is retained; detailed ownership of recruitment, scoring, analysis, and interpretation must be confirmed by the students from their notebooks. School psychologists are acknowledged for supplying observation reference material, and open-source authors are credited in the references. No funding or conflict declaration is invented from absent repository information.

This is an AI-assisted working draft for author review and independent rewriting, not a certified competition submission. AI assisted repository review, prose drafting, and build preparation. The authors must verify each reference, research claim, contribution, and applicable submission rule. School, city, and state identifiers and private participant images are omitted. No poster was prepared.

11 Conclusion

PSYCON provides an implemented pathway for studying contextual behavior with source-linked psychologist observations, distinct opportunity states, reviewed attribution, and independent evaluation boundaries. Retained engineering results demonstrate profile coverage and workflow execution. They do not answer the human behavioral hypotheses. The next scientific contribution depends on version-consistent, consented, independently reviewed observations and a held-out analysis that can support or challenge the proposed interpretation.

A Engineering reproducibility and historical work

The active runtime uses Flask/Jinja/plain JavaScript, a mandatory PostgreSQL store, a media worker, and an independent CPU trainer/import worker. PostgreSQL retains jobs, annotations, bounded source attachments, snapshots, metrics, and manifests; original media and model files remain registered local assets. A consistent backup requires both. Docker publishes the local app at port 8008; standalone web defaults to 8001. The image depends on a prebuilt speech runtime and authorized model access.

Jobs use row-locked claims, renewable leases, bounded retry, GPU-resource leasing, and owner/attempt/input-revision completion fencing. Revised inputs cannot be overwritten by a stale worker. Generated claims retain prompts, digest, decoding options, and source IDs; missing or mismatched local models fail explicitly. Same-origin mutation checks and loopback binding support a trusted local instrument, not public multi-user authorization.

Optional Executive, Builder, Salesperson, and Negotiator references use project-defined normalized marker distances. Their centers and similarity scores are exploratory and lack occupational ground truth. They cannot supply training labels or personality identity. The older bearer-token coaching, audiovisual, and physiology modules remain separate code paths and do not define the present BEHA conclusions.

The archived WESAD feature table contains 13,698 windows from fifteen subjects: 8,760 calm and 4,938 stress. Saved EDA-only XGBoost accuracy is 0.932, compared with 0.915 for multimodal logistic regression. These are a separate physiological task [16], and the artifacts do not support a blanket advantage from adding modalities. Calm baseline differences are computed using labelled subject windows before splitting, including held-out-subject calm information; calibration is not restricted to a pre-test interval. The scores therefore represent a label-informed calibration condition and cannot establish untouched unseen-person or PSYCON wearable performance.

Source traceability for the latest audit is provided by `docs/report_assets/maximum-defensible-summary.json` and `docs/research/maximum-defensible-2026-10-13/`. Earlier reported values are provided by `docs/report_assets/source-cross-validation-summary.json`, `docs/report_assets/source-training-summary.json`, `docs/report_assets/nemotron-benchmark-summary.json`, `docs/group_discussions_final_report.md`, `docs/INSTRUMENT_STATUS.md`, `results/metrics.json`, and the associated processing/training sources. The companion build script regenerates aggregate figures and both PDFs without accessing participant recordings or connecting to the canonical database.

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